Automated test log

Musanze Harvest Intelligence Service: pytest results

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Registration number	25/27389
Verification code	**AI-A2-7389**
Date	16 October 2026
Environment	Windows, Python 3.13, pytest
Command	`python -m pytest -q` (run from the project folder AI_A2_2527389)
Test file	tests/test_system.py

Total tests	Passed	Failed	Warnings	Duration
19	**19**	**0**	1 (harmless)	19.20 s

Terminal result as reported on the student's computer:

```
.................. [100%]
19 passed, 1 warning in 19.20s
```

The single warning is a StarletteDeprecationWarning from the library (use of httpx with the Starlette test client). It does not affect any test.

Test-by-test breakdown

No.	Test	Path	What it checks	Result
1	`test_valid_inference_returns_every_field`	Success	Valid record returns code, yield, probability, cluster and the human-review flag	**PASSED**
2	`test_missing_field_is_rejected`	Failure	Record without fertilizer_kg is rejected and the error names the field	**PASSED**
3	`test_wrong_type_is_rejected`	Failure	Text given for rainfall_mm is rejected	**PASSED**
4	`test_range_and_category_errors_are_rejected [rainfall_mm-1400.01]`	Failure	Rainfall just above the 1400 limit is rejected	**PASSED**
5	`test_range_and_category_errors_are_rejected [avg_temp_c-11.9]`	Failure	Temperature below the 12 limit is rejected	**PASSED**
6	`test_range_and_category_errors_are_rejected [field_size_ha-0.0]`	Failure	Field size below the 0.2 limit is rejected	**PASSED**
7	`test_range_and_category_errors_are_rejected [season-C]`	Failure	Season other than A or B is rejected	**PASSED**
8	`test_range_and_category_errors_are_rejected [cooperative_support-2]`	Failure	Support value other than 0 or 1 is rejected	**PASSED**
9	`test_unknown_field_is_rejected`	Failure	Extra field (yield_tons) is rejected, which blocks leakage through the API	**PASSED**
10	`test_wrong_seed_is_rejected`	Failure	A seed that is not the last four digits (7389) is refused	**PASSED**
11	`test_reproducibility_same_fingerprint_and_metrics`	Success	Two full pipeline runs give the same SHA-256 fingerprint and metrics	**PASSED**
12	`test_generator_row_count_follows_seed_rule`	Success	Base rows equal 240 + (7389 mod 41) = 249, plus 3 duplicates	**PASSED**
13	`test_deliberate_data_problems_are_detected_and_cleaned`	Success	12+12 missing values and 3 duplicates are detected and cleaned	**PASSED**
14	`test_leakage_guard_blocks_target_columns`	Failure	Adding high_risk to the features raises a validation error	**PASSED**

No.	Test	Path	What it checks	Result
15	`test_api_health_and_model_info`	Success	/health returns ok; /model-info shows AI-A2-7389 and the synthetic-data flag	**PASSED**
16	`test_api_predict_valid_response`	Success	POST /predict returns 200 with all required fields	**PASSED**
17	`test_api_invalid_request_gives_422_with_field_names`	Failure	Invalid POST returns 422 naming rainfall_mm and season	**PASSED**
18	`test_api_without_models_returns_503`	Failure	API with no trained models returns 503 instead of crashing	**PASSED**
19	`test_cli_rejects_invalid_record_with_nonzero_exit`	Failure	predict.py exits with code 2 and names the bad field	**PASSED**

Coverage: 13 of the 19 tests exercise failure paths (invalid input, wrong seed, leakage, missing models), and the rest cover successful behaviour and reproducibility. Test cases 4 to 8 are one parametrised test run with five different invalid inputs, which is why 15 test functions produce 19 results.